

Software Requirements Specification (SRS)

To-Do Web Application

Prepared by:- Aditya Mathur

Company :-Datansh

Date:-1/08/2026

Table of Contents

1. Introduction

➤ 1.1 Purpose

➤ 1.2 Scope

2. Overall Description

➤ 2.1 Product Functions

➤ 2.2 Users

3. Functional Requirements

4. Non-Functional Requirements

5. System Requirements

➤ 5.1 Hardware Requirements

➤ 5.2 Software Requirements

➤ 5.3 System Architecture

6. Conclusion

Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the requirements for a To-Do task management System. This document serves as a reference for developers and client during the design and development of the system.

The system helps user to assign tasks, track task progress, manage deadlines.

1.2 Scope

The Smart To-Do & Task Management System is a web-based application that allows users to organize and manage their daily tasks efficiently.

The system allows administrators to assign tasks, monitor progress, and communicate with users. Users can accept or decline assigned tasks, update task progress, mark tasks as completed, and ask questions related to tasks.

An AI chatbot is included to provide quick assistance and answer user queries.

The system enables:

- User registration and authentication
- Task creation, assignment, and management
- Task acceptance or rejection by users
- Task progress tracking and status updates
- Deadline and priority management
- AI chatbot assistance for answering user queries and providing productivity suggestions
- Notifications for new tasks, deadlines, and task updates
- Dashboard for monitoring assigned, pending, and completed tasks

OVERALL DESCRIPTION

2.1 Product Functions:-

➤ User Authentication

- **Register**
- **Login**
- **Logout**
- **Forgot Password**

➤ Task Management

- **Create Task**
- **Assign Task**
- **Edit Task**
- **Delete Task**
- **View Task Details**

➤ Task Status

- **Pending**
- **Accepted**
- **Declined**
- **In Progress**
- **Completed**

- **Progress Tracking**
 - Update completion percentage
 - View overall progress
 - Deadline tracking
- **Communication**
 - Ask questions to Admin
 - Admin replies to users
 - Task comments
- **Notifications**
 - New task assigned
 - Deadline reminder
 - Task status update
 - Completion notification
- **AI Chatbot**
 - Answer common questions
 - Explain task details
 - Help users navigate the application
 - Suggest productivity tips
- **Dashboard**
 - View today's tasks
 - Upcoming deadlines
 - Completed tasks
 - Pending tasks

2.2 USERS:-

➤ Administrator:-

The Administrator has complete control over the system and have these responsibilities:

- **Manage Users** – Add, update, or remove user accounts.
- **Create Tasks** – Create new tasks with priorities and deadlines.
- **Assign Tasks** – Assign tasks to one or more users.
- **Edit Tasks** – Modify task details whenever required.
- **Delete Tasks** – Remove tasks that are no longer needed.
- **Monitor Progress** – Track the status and completion of assigned tasks.
- **Generate Reports** – View reports on task completion, pending tasks, and user performance.

➤ User:-

The User performs the assigned tasks and can use the following features:

- **View Assigned Tasks** – Access all tasks assigned by the administrator.
- **Accept or Decline Tasks** – Choose whether to accept or reject a newly assigned task.
- **Update Progress** – Change the task status and update the completion percentage.
- **Mark Task as Completed** – Indicate when a task has been finished.
- **Use AI Chatbot** – Get help with task-related questions, application navigation, and productivity suggestions.
- **Receive Notifications** – Get updates about new tasks, deadlines, and task changes.

3. Functional Requirements:-

The TO-DO System shall provide the following functional requirements:

I. User Authentication

- **User Registration** – New users can create an account by providing their name, email address, and password.
- **User Login** – Registered users can securely log in using their email and password.
- **Forgot Password** – Users can reset their password through their registered email.
- **Logout** – Users can securely log out of the application.

II. Task Management

- **Create Task** – The administrator can create a new task by providing priority and deadline.
- **Assign Task** – The administrator can assign tasks to one or more users.
- **Edit Task** – The administrator can modify task details such as title, description, priority, deadline, or assigned user.
- **Delete Task** – The administrator can remove tasks that are no longer required.

III. Task Handling

- **View Assigned Tasks** – Users can view all tasks assigned to them.
- **Accept Task** – Users can accept a newly assigned task to begin working on it.
- **Decline Task** – Users can decline a task if they are unable to complete it. The administrator will be notified.

- **Update Task Status** – Users can update the task status as Pending, Accepted, In Progress, Completed, or Declined.
- **Update Progress** – Users can update the completion percentage of a task.
- **Mark Task as Completed** – Users can mark a task as completed after finishing the assigned work.

IV. Communication

- **Ask Questions** – Users can send queries to the administrator regarding any assigned task.
- **Reply to Queries** – The administrator can respond to user questions directly through the application.
- **Task Comments** – Both administrators and users can add comments to a task for better communication and tracking.

V. AI Chatbot

- **Task Assistance** – The AI chatbot can explain task details and help users understand their assigned work.
- **Application Guidance** – The chatbot can assist users in navigating different features of the application.
- **Frequently Asked Questions** – The chatbot can answer common questions related to the system.
- **Productivity Suggestions** – The chatbot can provide simple tips for better task planning and time management.

VI. Notifications

- **Task Assignment Notification** – Users receive a notification whenever a new task is assigned.
- **Deadline Reminder** – The system reminds users before a task deadline.

- **Task Update Notification** – Users are notified whenever a task is modified by the administrator.
- **Query Response Notification** – Users receive a notification when the administrator replies to their query.
- **Task Completion Notification** – The administrator is notified when a user marks a task as completed.

VII. Dashboard

Administrator Dashboard

- **Task Overview** – View the total number of tasks in the system.
- **Progress Monitoring** – Monitor the status and progress of all assigned tasks.
- **User Performance** – View task completion statistics for each user.
- **Recent Activities** – Track recently assigned, updated, and completed tasks.

User Dashboard

- **Assigned Tasks** – View all assigned tasks in one place.
- **Task Summary** – View pending, in-progress, completed, and overdue tasks.
- **Upcoming Deadlines** – View tasks that are approaching their due dates.
- **Notifications** – Access recent task updates and reminders.

VIII. Reports

- **Task Reports** – The administrator can generate reports showing completed, pending, and overdue tasks.
- **User Performance Reports** – View task completion statistics and productivity of individual users.

- **Progress Reports** – Monitor overall project progress through generated reports.

IX. Search and Filter

- **Search Tasks** – Users can search for tasks using keywords or task titles.
- **Filter Tasks** – Users can filter tasks based on status, priority, assigned user, or due date.
- **Sort Tasks** – Users can sort tasks by deadline, priority, or creation date.

X. Profile Management

- **Update Profile** – Users can edit their personal information.
- **Change Password** – Users can securely update their account password.
- **Profile Picture** – Users can upload or change their profile picture.

4. Non-Functional Requirements:-

The Smart Task Management System shall satisfy the following non-functional requirements:

Performance

- **Fast Response Time** – The system should load pages and process requests within 2–3 seconds under normal usage.
- **Concurrent Users** – The system should support multiple users accessing the application at the same time without significant performance issues.

Security

- **User Authentication** – Only registered users should be able to access the system through secure login.
- **Data Protection** – User passwords should be encrypted, and personal information should be stored securely.
- **Role-Based Access** – Administrators and users should only have access to features permitted by their roles.

Reliability

- **Data Integrity** – Task information should be stored accurately without data loss.
- **System Stability** – The application should operate consistently with minimal errors or crashes.

Availability

- **System Uptime** – The application should be available 24×7 except during scheduled maintenance.

Usability

- **User-Friendly Interface** – The system should provide a simple and easy-to-use interface for all users.

- **Easy Navigation** – Users should be able to access features with minimal effort and clear navigation.

Scalability

- **Future Growth** – The system should support an increasing number of users and tasks without major changes to the application.

Maintainability

- **Easy Updates** – The system should be designed so that new features, bug fixes, and improvements can be implemented easily.

Compatibility

- **Cross-Platform Support** – The application should work on major web browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox, and be accessible on desktops and laptops.

5. Technical Specifications

5.1 Hardware Requirements

The application requires the following minimum hardware specifications for smooth operation.

Client System

- **Processor** – Intel Core i3 or higher
- **RAM** – Minimum 4 GB
- **Storage** – At least 10 GB of free disk space
- **Internet Connection** – Stable broadband connection

Server System

- **Processor** – Quad-Core Processor or higher
- **RAM** – Minimum 8 GB
- **Storage** – SSD with sufficient capacity for application and database
- **Network** – High-speed internet connection

5.2 Software Requirements

The following software technologies are required for developing and running the application.

Frontend

- **React.js** – Used to build the user interface.
- **TypeScript** – Improves code quality and maintainability.
- **HTML5** – Provides the structure of web pages.
- **CSS3** – Used for styling and responsive design.

Backend

- **Java** – Primary programming language used for backend development.
- **Spring Boot** – Used to build RESTful APIs, implement business logic, and manage backend services.
- **Spring Security** – Provides secure authentication and role-based authorization.
- **Spring Data JPA** – Simplifies database operations and kJnm communication with PostgreSQL.
- **Maven** – Used for dependency management and project build automation.

Database

- **PostgreSQL** – Stores user information, task details, comments, notifications, and application data.

AI Integration

- **AI API** – Provides chatbot functionality for answering user queries and assisting with task-related information.

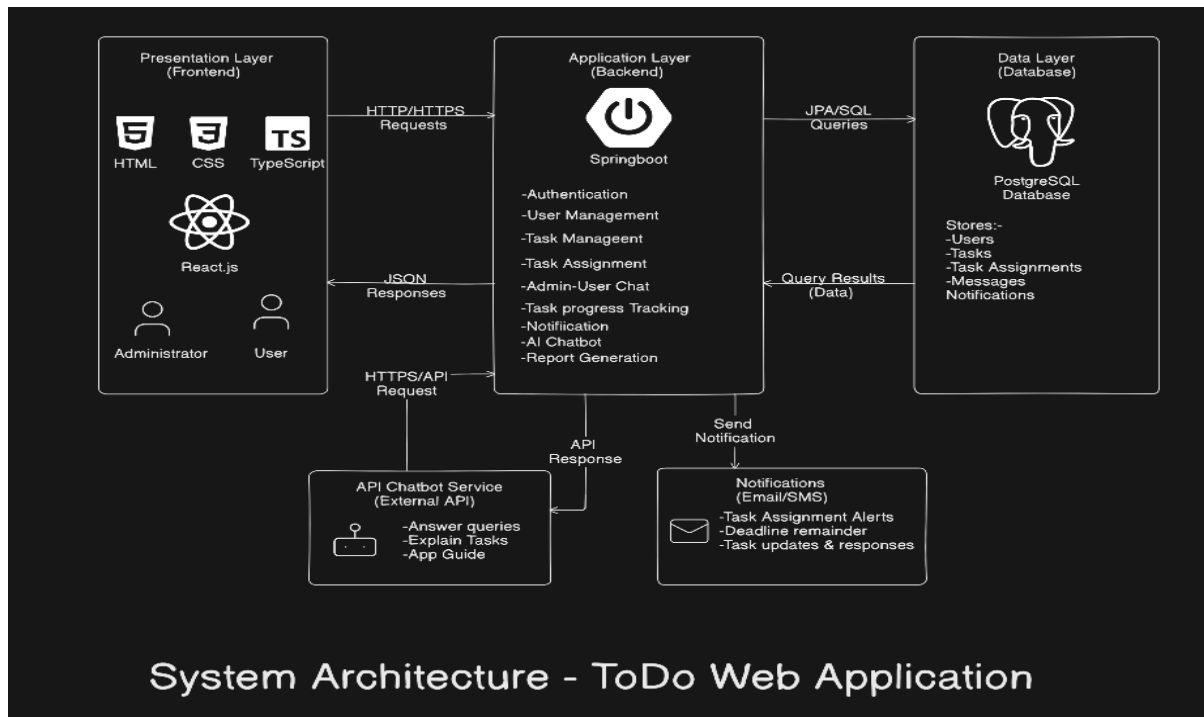
Development Tools

- Visual Studio Code
- Git
- GitHub
- Postman

Supported Browsers

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

5.3 System Architecture



6. Conclusion

This Software Requirements Specification (SRS) document outlines the functional and non-functional requirements of the Smart Task Management System. It provides a clear understanding of the system's objectives, features, technical requirements, and user responsibilities.

The application is designed to simplify task management by enabling administrators to assign and monitor tasks while allowing users to manage their assigned work efficiently. Features such as task progress tracking, notifications, communication between administrators and users, and an AI chatbot enhance the overall user experience and improve productivity.

This document serves as a reference for the design, development, testing, and maintenance of the application, ensuring that the system is developed according to the specified requirements.